

Installation and User Guide

1. Introduction

This guide provides instructions to install and use the JTP Product Recommendation System, a full-stack application featuring a React frontend, FastAPI backend, and PostgreSQL database with vector search capabilities.

2. Prerequisites

- Docker
- Git
- VSCode (recommended)

3. Installation

1. Clone the Repository:

```
...  
git clone https://github.com/1aryantyagi/JTP-project  
cd JTP-project  
...
```

2. Run Docker Compose:

```
...  
docker-compose up --build  
...
```

- This will build and start:

- Backend (FastAPI) on port 8000
- Frontend on port 3000
- PostgreSQL database with pgvector on port 5432

4. Accessing the Application

- Frontend: <http://localhost:3000> (THIS IS MAIN)
- Backend API Docs: <http://localhost:8000/docs>
- Database: Accessible inside the `postgres` container via port 5432

5. Using the Application

Main Page

- When you first open the application at <http://localhost:3000>, you land on the **Main Page**.
- This page displays a random selection of products pulled from the database.
- You can click the **“Refresh Products”** button at the top to reload and view a new set of random products.
- Each product is shown with its name, image, brand, price, and buttons for interaction.

Product Recommendation

- Next to each product is a **“View Recommendation”** button.
- Clicking this sends a request to the backend to find **up to 7 similar items** based on product embeddings and cosine similarity using pgvector.
- The results are displayed on a separate **Recommendation Page**.
- For example, clicking "View Recommendation" on **Digestive Biscuits** will show other biscuit types or similar snacks.

Adding to Cart

- To add a product to your cart, click the **“Add to Cart”** button below the product.
- This adds the item to your personal cart stored on the frontend.
- The cart icon in the navigation bar (top right) updates with the number of items.

Cart Page

- Click the cart icon to go to the **Cart Page**, which lists all products added by the user.
- You can remove products from the cart or proceed to recommendations based on the cart's contents.
- At the bottom, there is a **“Frequently Bought Together”** section.
 - This section recommends products that are often purchased along with the items currently in your cart.
 - It uses a frequency-based aggregation method over vector similarity searches for all cart items.

Sorting Products

- On the Main Page, you will find a **“Sort by Price”** dropdown.
- You can use this to reorder all displayed products by their **sale price** in ascending or descending order.
- This helps users find affordable or premium items more quickly, enhancing user experience.

6. API Endpoints (For Developers)

- **GET /random_products:** Returns a random selection of products
- **GET /recommend/{product_name}:** Returns up to 7 products similar to `product\_name` based on vector similarity.
- **POST /recommend/cart** Accepts a JSON payload with cart items and returns up to 12 aggregated product recommendations.

7. Stopping the Application

To stop services, press Ctrl+C in the terminal running Docker Compose, then run:

```
...
```

```
docker-compose down -v
```

```
...
```

8. Troubleshooting

Build Failures Ensure Docker is running and ports 3000, 8000, and 5432 are available.

Database Initialization If the database does not initialize correctly, check the logs of the `postgres` container for errors.

Frontend Not Loading Verify the backend is online at <http://localhost:8000> and healthy.